Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\cos^{-1} x + \cos^{-1}(2/3) = (\pi/2)$, the value of x is

(A) $1/3$

(B) $\sqrt{5}/3$

(C) $\sqrt{5}/2$

(D) $1/2$

Q.2

$\sin^{-1}(4/5) + \sin^{-1}(5/13) + \sin^{-1}(16/65)$ is equal to

(A) 0

(B) $\pi/2$

(C) π

(D) $\sin^{-1}(63/65)$

Q.3

If $a_1, a_2, a_3, \ldots\ldots a_n$ is an A.P. with common difference d; $(d > 0)$ then
$\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + \ldots\right.$
$\left.\ldots\ldots + \tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right]$ is equal to

(A) $\frac{(n-1)d}{a_1+a_n}$

(B) $\frac{(n-1)d}{1+a_1a_n}$

(C) $\frac{nd}{1+a_1a_n}$

(D) $\frac{a_n-a_1}{a_n+a_1}$

Q.4

If $\cos^{-1} x + \cos^{-1} y = \frac{2\pi}{7}$, then the value $\sin^{-1} x + \sin^{-1} y$ is equal to

(A) $\frac{4\pi}{7}$

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(B) $\frac{3\pi}{7}$

(C) $\frac{2\pi}{7}$

(D) $\frac{5\pi}{7}$

Q.5

If $\sin^{-1} a + \sin^{-1} b + \sin^{-1} c = \pi$, then find the value of $a\sqrt{1-a^2} + b\sqrt{1-b^2} + c\sqrt{1-c^2}$

(A) abc

(B) $a + b + c$

(C) $\frac{1}{a} \times \frac{1}{b} \times \frac{1}{c}$

(D) $2abc$

Q.6

If $\sin^{-1}x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots \infty + \cos^{-1}x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots \infty = \frac{\pi}{2}$ and $0 < x < \sqrt{2}$, then $x =$

(A) $\frac{1}{2}$

(B) 1

(C) $-\frac{1}{2}$

(D) -1

Q.7

In a ΔABC, if $A = \tan^{-1} 2$ and $B = \tan^{-1} 3$, then $C =$

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{3\pi}{4}$

Q.8

If $x \in [\pi/2, \pi]$ then $\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right) =$

(A) $\frac{x-\pi}{2}$

(B) $\frac{\pi-x}{2}$

(C) $\frac{3\pi-x}{2}$

(D) None of these

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Q.9

If $u = \cot^{-1}\sqrt{\tan\alpha} - \tan^{-1}\sqrt{\tan\alpha}$, then $\tan\left(\frac{\pi}{4} - \frac{u}{2}\right)$ is equal to

(A) $\sqrt{\tan\alpha}$

(B) $\sqrt{\cot\alpha}$

(C) $\tan\alpha$

(D) $\cot\alpha$

Q.10

The number of roots of the equation $\sin^{-1}x - \cos^{-1}x = \sin^{-1}5x - 3$ is/are

(A) 3

(B) 1

(C) 2

(D) 0

Q.11

Solving $2\cos^{-1}x = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$, we get

(A) $x \in \left[\frac{\sqrt{2}}{2}, 1\right]$

(B) $x = 3$

(C) $x \in [3, 4]$

(D) $x = 0$

Q.12

If $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$ then x is equal to

(A) -1

(B) -2

(C) 1

(D) 2

Q.13

If $\sin^{-1}x + \sin^{-1}y = \frac{\pi}{2}$, then $\frac{1+x^4+y^4}{x^2-x^2y^2+y^2}$ is equal to

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(A) 1

(B) 2

(C) $1/2$

(D) None of these

Q.14

The value of $\cot^{-1} 7 + \cot^{-1} 8 + \cot^{-1} 18$ is

(A) π

(B) $\frac{\pi}{2}$

(C) $\cot^{-1} 5$

(D) $\cot^{-1} 3$

Q.15

$\sec^2 \left(\tan^{-1} 2\right) + \text{cosec}^2 \left(\cot^{-1} 3\right) =$

(A) 5

(B) 13

(C) 15

(D) 6

Answers & Solutions

Q.1 Answer:

$\sqrt{5}/3$

Solution:

$$\cos^{-1} x = \pi/2 - \cos^{-1} 2/3$$
$$\Rightarrow \quad x = \cos\left(\frac{\pi}{2} - \cos^{-1}\frac{2}{3}\right) = \sin\cos^{-1}\left(\frac{2}{3}\right)$$
$$= \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$$

Q.2 Answer:

$\pi/2$

Solution:

$$\sin^{-1}\left[(4/5)\sqrt{1-(5/13)^2} + \tfrac{5}{13}\sqrt{1-(4/5)^2}\right] + \sin^{-1}(16/65)$$
$$= \sin^{-1}[(4/5)\times(12/13) + (5/13)\times(3/5)] + \sin^{-1}(16/65)$$

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$= \sin^{-1}(63/65) + \sin^{-1}(16/65)$

$= \cos^{-1}\sqrt{1-(63/65)^2} + \sin^{-1}(16/65)$

$= \cos^{-1}(16/65) + \sin^{-1}(16/65) = \pi/2$

Q.3 Answer:

$\frac{(n-1)d}{1+a_1a_n}$

Solution:

$a_1, a_2, \ldots, a_n$ **are in A.P with common difference** d**.**

$a_2 = a_1 + d$

$a_3 = a_1 + 2\,d$

$a_n = a_1 + (n-1)d$

$\tan^{-1}\left(\frac{d}{1+a_1a_2}\right) = \tan^{-1}\left(\frac{a_2-a_1}{1+a_1a_2}\right) = \tan^{-1}(a_2) - \tan^{-1}(a_1)$

$\tan^{-1}\left(\frac{d}{1+a_2a_3}\right) = \tan^{-1}\left(\frac{a_3-a_2}{1+a_2a_3}\right) = \tan^{-1}(a_3) - \tan^{-1}(a_2)$

$\tan^{-1}\left(\frac{d}{1+a_n+a_n}\right) = \tan^{-1}\left(\frac{a_n-a_{n-1}}{1+a_{n-1}a_n}\right) = \tan^{-1}(a_n) - \tan^{-1}(a_{n-1})$

$\therefore \tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + .. + \tan^{-1}\left(\frac{d}{1+a_n+a_n}\right)$

$= \tan^{-1}(a_n) - \tan^{-1}(a_1)$

$= \tan^{-1}\left(\frac{a_n-a_1}{1+a_1a_n}\right)$

$= \tan^{-1}\left(\frac{a_1+(n-1)d-a_1}{1+a_1a_n}\right)$

$= \tan^{-1}\left(\frac{(n-1)d}{1+a_1a_n}\right)$

$\therefore \tan\left\{\tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + \ldots + \tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right\}$

$= \tan\left\{\tan^{-1}\left(\frac{(n-1)d}{1+a_1a_n}\right)\right\}$

$= \frac{(n-1)d}{1+a_1a_n}$

Q.4 Answer:

$\frac{5\pi}{7}$

Solution:

Let $\cos^{-1}x + \cos^{-1}y = \frac{2\pi}{7}$

$\Rightarrow \left(\frac{\pi}{2} - \sin^{-1}x\right) + \left(\frac{\pi}{2} - \sin^{-1}y\right) = \frac{2\pi}{7}$

$\Rightarrow \sin^{-1}x + \sin^{-1}y = \pi - \frac{2\pi}{7} = \frac{5\pi}{7}.$

Q.5 Answer:

$2abc$

Solution:

Let $\sin^{-1}a = x \quad \therefore a = \sin x$

$\sin^{-1}b = y \quad \therefore b = \sin y$

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$\sin^{-1} c = z \quad \therefore c = \sin z$

$\therefore \quad a\sqrt{1-a^2} + b\sqrt{1-b^2} + c\sqrt{1-c^2}$

$= \sin x \cos x + \sin y \cos y + \sin z \cos z$

$= (1/2)(\sin 2x + \sin 2y + \sin 2z) = (1/2)(4 \sin x \sin y \sin z)$

$= 2 \sin x \sin y \sin z = 2abc$

Q.6 Answer:

1

Solution:

As we know $\sin^{-1}x + \cos^{-1}y = \frac{\pi}{2}$ **only when** $x = y$

$\Rightarrow x - \frac{x^2}{2} + \frac{x^3}{4} \ldots\ldots\infty = x^2 - \frac{x^4}{2} + \frac{x^6}{4} \ldots\ldots\infty$

$\Rightarrow \frac{x}{1 - -\frac{x}{2}} = \frac{x^2}{1 - -\frac{x^2}{2}}$

$\Rightarrow x = 1$

Q.7 Answer:

$\frac{\pi}{4}$

Solution:

We have, $A = \tan^{-1} 2 \Rightarrow \tan A = 2$

and $\mathrm{B} = \tan^{-1} 3 \Rightarrow \tan \mathrm{B} = 3$

since A, B, C **are angles of a triangle,** $A + B + C = \pi \Rightarrow \mathrm{C} = \pi - (\mathrm{A} + \mathrm{B})$

Now, $\mathrm{A} + \mathrm{B} = \tan^{-1} 2 + \tan^{-1} 3 = \pi + \tan^{-1}\left[\frac{2+3}{1-2.3}\right]$

$\left[\begin{array}{l} \therefore \tan^{-1} \mathrm{x} + \tan^{-1} \mathrm{y} = \pi + \tan^{-1}\left[\frac{\mathrm{x}+\mathrm{y}}{1-\mathrm{xy}}\right] \\ \text{for } \mathrm{x} > 0, \mathrm{y} > 0 \text{ and } \mathrm{xy} > 1 \end{array}\right.$

$= \pi + \tan^{-1}(-1) = \pi - \tan^{-1} 1 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$\therefore$ **From** $(1), \mathrm{C} = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$

Q.8 Answer:

$\frac{\pi - x}{2}$

Solution:

$\sqrt{1 + \sin x} = \sin\frac{x}{2} + \cos\frac{x}{2}$

$\sqrt{1 - \sin x} = \sin\frac{x}{2} - \cos\frac{x}{2}$

$\left[\text{ for } \frac{\pi}{4} \leq \frac{x}{2} \leq \frac{\pi}{2} \sin\frac{x}{2} \geq \cos\frac{x}{2}\right]$

$\therefore$ **the expression is** $\cot^{-1}\left(\frac{\sin\frac{x}{2} + \cos\frac{x}{2} + \sin\frac{x}{2} - \cos\frac{x}{2}}{\sin\frac{x}{2} + \cos\frac{x}{2} - \sin\frac{x}{2} + \cos\frac{x}{2}}\right)$

$= \cot^{-1}\left(\tan\frac{x}{2}\right) = \cot^{-1}\cot\left(\frac{\pi}{2} - \frac{x}{2}\right) = \frac{\pi - x}{2}$

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Q.9 Answer:

$\sqrt{\tan\alpha}$

Solution:

Let $\sqrt{\tan\alpha} = \tan x$,
then $u = \cot^{-1}(\tan x) - \tan^{-1}(\tan x)$
$= \frac{\pi}{2} - x - x = \frac{\pi}{2} - 2x$
$\Rightarrow 2x = \frac{\pi}{2} - u \Rightarrow x = \frac{\pi}{4} - \frac{u}{2}$
$\Rightarrow \tan x = \tan\left(\frac{\pi}{4} - \frac{u}{2}\right) \Rightarrow \sqrt{\tan\alpha} = \tan\left(\frac{\pi}{4} - \frac{u}{2}\right)$

Q.10 Answer:

1

Solution:

$\sin^{-1}x - \cos^{-1}x = \sin^{-1}5x - 3$
$\Rightarrow \frac{\pi}{2} - \cos^{-1}x - \cos^{-1}x = \frac{\pi}{2} - \cos^{-1}5x - 3$
$\Rightarrow 2\cos^{-1}x = \cos^{-1}5x - 3$. Also $x \in -1,1$ **...**
1
$\Rightarrow \cos^{-1}2x^2 - 1 = \cos^{-1}5x - 3$ **and**
$5x - 3 \in -1,1$, **i.e.,** $-1 \le 5x - 3 \le 1$
$\Rightarrow 2x^2 - 1 = 5x - 3$, **hence** $x \in \frac{2}{5}, \frac{4}{5}$
$\Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow x = 2$ **or** $\frac{1}{2}$
but, $x = 2$ **does not satisfy the equation** 1
Hence, the given equation has only one root

Q.11 Answer:

$x \in \left[\frac{\sqrt{2}}{2}, 1\right]$

Solution:

$x = \cos y$; **where** $0 \le y \le \pi, |x| \le 1$
$2\cos^{-1}x = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$
$\Rightarrow 2\cos^{-1}(\cos y) = \sin^{-1}\left(2\cos y \cdot \sqrt{1-\cos^2 y}\right)$
$\Rightarrow 2\cos^{-1}(\cos y) = \sin^{-1}(2\cos y \cdot \sin y)$
$\Rightarrow 2\cos^{-1}(\cos y) = \sin^{-1}(\sin 2y)$
$\Rightarrow \sin^{-1}(\sin 2y) = 2y$ **for** $-\pi/4 \le y \le \pi/4$
and $2\cos^{-1}(\cos y) = 2y$ **for** $0 \le y \le \pi$
Thus, Eq. (i) holds only when, $y \in [0, \pi/4] \quad \Rightarrow x \in [\sqrt{2}/2, 1]$

Q.12 Answer:

-1

Solution:

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Given : $\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}$
$\Rightarrow \tan^{-1}\frac{(2x+3x)}{(1-2x.3x)}=\tan^{-1}(1)$
$\Rightarrow \frac{5x}{1-6x^2}=1 \quad \Rightarrow \quad 6x^2+5x-1=0$
$\Rightarrow (6x-1)(x+1)=0 \Rightarrow x=\frac{1}{6}$ **or** -1

Q.13 Answer:

2

Solution:

$\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2} \Rightarrow \sin^{-1}x=\frac{\pi}{2}-\sin^{-1}y$
$\Rightarrow \sin^{-1}x=\sin^{-1}\sqrt{1-y^2} \Rightarrow x^2+y^2=1$
$\Rightarrow \frac{1+x^4+y^4}{x^2-x^2y^2+y^2}=\frac{1+\left(x^2+y^2\right)^2-2x^2y^2}{1-x^2y^2}$
$=\frac{1+1-2x^2y^2}{1-x^2y^2}=2$

Q.14 Answer:

$\cot^{-1}3$

Solution:

We have $\cot^{-1}7+\cot^{-1}8+\cot^{-1}18$
$\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}+\tan^{-1}\frac{1}{18}$
$=\tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right)+\tan^{-1}\frac{1}{18}$
$=\tan^{-1}\frac{15}{55}+\tan^{-1}\frac{1}{18}\left(\because \frac{1}{7}\cdot\frac{1}{8}<1\right)$
$\tan^{-1}\frac{3}{11}+\tan^{-1}\frac{1}{18}=\tan^{-1}\left(\frac{\frac{3}{11}+\frac{1}{18}}{1-\frac{3}{11}\times\frac{1}{18}}\right)$
$\left(\because \frac{3}{11}\cdot\frac{1}{18}<1\right)$
$=\tan^{-1}\frac{65}{195}=\tan^{-1}\frac{1}{3}=\cot^{-1}3$

Q.15 Answer:

15

Solution:

Let $\tan^{-1}2=a \Rightarrow \tan\alpha=2$
and $\cot^{-1}3=\beta \Rightarrow \cot\beta=3$
$\sec^2\left(\tan^{-1}2\right)+\operatorname{cosec}^2\left(\cot^{-1}3\right)$
$=\sec^2\alpha+\operatorname{cosec}^2\beta=1+\tan^2\alpha+1+\cot^2\beta$
$=2+(2)^2+(3)^2=15$

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